

CST8506
ADVANCED
MACHINE LEARNING

Week 12
Misc Topics

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Agenda

- Loss Functions
- Parameters in NN algorithms



Loss Functions

- Method to evaluate how well an algorithm models the given data
- Quantifies the error between the output and the target
- Also known as cost function or error function
 - Regression Losses
 - Probabilistic Losses
 - Hinge Losses for maximum-margin classification
- <https://keras.io/api/losses/>



Regression Loss Functions

- Mean Square Error (MSE) / Quadratic Loss / L2 Loss
 - Average of the sum of the squared differences between actual value and the predicted value
- Mean Absolute Error (MAE) / L1 Loss
 - Average of the sum of the absolute differences between actual value and the predicted value
 - Robust to outliers since it does not make use of square
- Mean Bias Error
 - Average of the sum of the differences between actual value and the predicted value
 - Positive and negative values may cancel out – less accurate in practice
 - Can be used to see whether model has positive or negative bias



MSE & MAE

$$w = -266.5344 + 6.1376h$$

		x	y _i	y _i '	y _i - y _i '	(y _i - y _i ') ²
-266.53	6.1376	63	127	120.1388	6.8612	47.076065
-266.53	6.1376	64	121	126.2764	-5.2764	27.840397
-266.53	6.1376	66	142	138.5516	3.4484	11.891463
-266.53	6.1376	69	157	156.9644	0.0356	0.0012674
-266.53	6.1376	69	162	156.9644	5.0356	25.357267
-266.53	6.1376	71	156	169.2396	-13.2396	175.28701
-266.53	6.1376	71	169	169.2396	-0.2396	0.0574082
-266.53	6.1376	72	165	175.3772	-10.3772	107.68628
-266.53	6.1376	73	181	181.5148	-0.5148	0.265019
-266.53	6.1376	75	208	193.79	14.21	201.9241
					Total	597.38627

Prediction Error	Absolute Error	Squared Error	
6.8612	6.8612	47.07607	
-5.2764	5.2764	27.8404	
3.4484	3.4484	11.89146	
0.0356	0.0356	0.001267	
5.0356	5.0356	25.35727	
-13.2396	13.2396	175.287	
-0.2396	0.2396	0.057408	
-10.3772	10.3772	107.6863	
-0.5148	0.5148	0.265019	
14.21	14.21	201.9241	
Mean Absolute Error	5.92384	7.729077	Root Mean Squared Error
		59.73863	Mean Square Error (MSE)



Probabilistic Loss Functions

- Binary Cross Entropy (two classes – 0 and 1 as class labels)
- Categorical Cross Entropy (one-hot encoded class labels)
- Sparse Categorical Cross Entropy (integers as class labels)



Cross Entropy

What is Entropy?

Cross Entropy – measure of the difference between two probability distributions

- $CCE = \sum_{i=1}^C t_i \log(s_i)$
where t_i are the targets and s_i are the probabilities of t_i
- $BCE = \sum_{i=1}^2 t_i \log(s_i)$
 $= -t_1 \log(s_1) - (1 - t_1) \log(1 - s_1)$

Cross Entropy is also known as log loss.

https://www.youtube.com/watch?v=Pwgpl9mKars&ab_channel=AdianLiusie

https://www.youtube.com/watch?v=Md4b67HvmRo&ab_channel=DigitalSreeni

https://www.youtube.com/watch?v=6ArSys5qHAU&ab_channel=StatQuestwithJoshStarmer



Batch vs Iteration vs Epoch

Epoch: the number of times the whole training dataset is passed through the model.

Batch: An epoch can have number of batches.

Iteration: number of batches required to complete one epoch.

Example: if we have 3000 instances and batch size is 500 then we have $3000/500$ iterations per epoch

<https://www.kaggle.com/getting-started/193766>

